Local ordinary discriminants of Rogers Szegő polynomials at odd roots of unity

An all order derivation with separated and simple overlap hypotheses

Revised working mathematical note
Version 2 11 October 2026

1 Purpose and scope

We derive the first moving-root terms of Rogers–Szegő polynomials at a fixed odd root of unity, for every positive block multiplicity A. The finite q-binomial theorem gives an exact diagonal-operator identity. A block-divisibility lemma then controls every order of the formal remainder. In the case of one simple shared root, only one additional product derivative is needed; its coefficient reduces to the classical value at $z = -1$.

Under the stated simplicity and separation hypotheses, these initial forms determine the exact cyclotomic order and the complete leading coefficient of the ordinary discriminant. We retain all signs and distinguish the parameters $q - \omega$, $q^d - 1$, and $\Phi_d(q)$.

The all-order proof underlying this note received an independent scoped acceptance on 11 October 2026. This note reorganizes that accepted proof and its qualifications into a single mathematical account. Historical publication priority remains unresolved. The conclusion is an explicit all-order derivation from classical inputs; it does not assert that the displayed formulas were previously published or establish a worldwide originality claim. Source identities and the finite-control record appear in the appendices.

Version 2 adds Appendix C, a concise integration of the independently accepted finite-parameter analytic remainder, root-separation, and discriminant-error certificate. It supplies explicit inputs to the existing Q08 inverse-Vandermonde bound. The formal theorems and their proofs are unchanged. Acceptance of the source mathematics is distinct from independent document-fidelity review of this revision; Appendix D identifies the frozen sources.

2 Definitions and conventions

Let K be a characteristic-zero field containing a primitive dth root of unity ω, where $d \geq 3$ is odd. Fix integers $A \geq 1$ and $0 \leq b < d$, and put $n = Ad + b$. Algebraic closures and formal power-series extensions of K may be used to discuss individual roots.

The Gaussian binomial coefficient is the polynomial in $\mathbb{Z}[q]$ represented, for $0 \leq k \leq n$, by

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{j=1}^{k} \frac{1 - q^{n-k+j}}{1 - q^j}.$$

Specialization at a root of unity means evaluation of this polynomial, rather than substitution into an unreduced quotient. Define

$$H_n(z;q)=\sum_{k=0}^{n}\begin{bmatrix}n\\k\end{bmatrix}_q z^k,\qquad D_n(q)=\operatorname{Disc}_z H_n(z;q),$$

$$F(z)=1+z^d,\qquad G(z)=H_b(z;\omega),\qquad P_A=\prod_{j=1}^{A}j^j.$$

Every H_n is monic in z. For a monic polynomial $f(z)=\prod_i(z-r_i)$, the ordinary discriminant and the resultant with a monic polynomial g are

$$\operatorname{Disc}(f)=\prod_{i<j}(r_i-r_j)^2,\qquad \operatorname{Res}(f,g)=\prod_i g(r_i).$$

We use $\operatorname{Disc}(1)=\operatorname{Disc}(\text{monic linear})=1$ and empty products equal to 1. All discriminants below use the ordinary z derivative, not a q-difference operator.

The monic probabilists' Hermite polynomial is

$$\operatorname{He}_m(y)=m!\sum_{j=0}^{\lfloor m/2\rfloor}\frac{(-1)^j y^{m-2j}}{2^j j!(m-2j)!}.$$

In particular $\operatorname{He}_0=1$, $\operatorname{He}_1=y$, and $\operatorname{Disc}(\operatorname{He}_A)=P_A$. Write $(a;q)_m=\prod_{j=0}^{m-1}(1-aq^j)$.

We use formal local parameters

$$q=\omega e^h,\qquad q(\varepsilon)=\omega(1+\varepsilon^2)^{1/d},\qquad h=\frac{\log(1+\varepsilon^2)}{d}.$$

The binomial and logarithmic series have their constant terms fixed as written. Thus $q(0)=\omega$, $q(\varepsilon)^d-1=\varepsilon^2$, and $h=\varepsilon^2/d+O(\varepsilon^4)$. A remainder $O(\varepsilon^s)$ in a local polynomial identity is divisible by ε^s in $K[x][[\varepsilon]]$, after adjoining square roots if convenient. These square roots cancel from the displayed initial forms coefficientwise.

3 The two local theorems

3 1 Separated roots

Theorem 1. For every root α of F, the following coefficient identity holds:

$$H_n(\alpha+\varepsilon x;q(\varepsilon))=\varepsilon^A(-1)^A d^{A/2}G(\alpha)\operatorname{He}_A\left(\frac{\sqrt{d}\,x}{\alpha}\right)+O(\varepsilon^{A+1}). \tag{1}$$

If $G(\alpha)=0$, the written coefficient vanishes and (1) does not identify the first nonzero term. For the separated discriminant conclusion assume

$$\operatorname{Disc}(G)\operatorname{Res}(F,G)\neq 0. \tag{2}$$

Put $M_{\mathrm{sep}}=dA(A-1)/2$. Then

$$D_n(q(\varepsilon))=\varepsilon^{2M_{\mathrm{sep}}}(C_{\mathrm{sep}}+O(\varepsilon^2)), \tag{3}$$

$$C_{\mathrm{sep}} = (-1)^{A(d-1)/2} d^{dA(A+1)/2} P_A^d \operatorname{Disc}(G) \operatorname{Res}(F,G)^{2A}. \tag{4}$$

The coefficient is nonzero, and $\operatorname{ord}_{\Phi_d} D_n = M_{\mathrm{sep}}$. The same separated proof also covers the classical endpoint $d = 1$, $b = 0$; this endpoint is not presented as a new result.

3 2 One simple shared root

Suppose b is odd, so $1 \le b < d$, and write

$$G(z) = (z+1)U(z), \qquad g = U(-1) \neq 0.$$

Set

$$m = \frac{d+b}{2} < d, \qquad W = \frac{(\omega;\omega)_b}{(\omega^2;\omega^2)_m}, \qquad \tau = \frac{dW}{g}, \tag{5}$$

$$J_{A+1}(y;\tau) = \mathrm{He}_{A+1}(y) + A(1-\tau)\,\mathrm{He}_{A-1}(y). \tag{6}$$

Every denominator factor in W is nonzero: since d is odd, ω^2 also has order d, and $m < d$.

Theorem 2. Under $g \neq 0$, the shared-root coefficient identity is

$$H_n(-1+\varepsilon x; q(\varepsilon)) = \varepsilon^{A+1} d^{(A-1)/2} g J_{A+1}(\sqrt{d}\,x;\tau) + O(\varepsilon^{A+2}). \tag{7}$$

Both Hermite summands have parity $A+1$. Therefore the leading factor in (7) is equivalently $(-1)^{A+1} d^{(A-1)/2} g J_{A+1}(-\sqrt{d}\,x;\tau)$, as in the accepted source statement.

For the global overlap discriminant additionally assume

$$\operatorname{Disc}(U) \operatorname{Res}(F,U) \operatorname{Disc}(J_{A+1}) \neq 0. \tag{8}$$

Here $\operatorname{Disc}(J_{A+1})$ means the discriminant in y at the specified value of τ. These hypotheses give exactly one shared residual root, namely -1, simple in G, with distinct first displacements. In fact $\operatorname{Res}(F,U) \neq 0$ already implies $g \neq 0$.

Put $M_{\mathrm{ov}} = dA(A-1)/2 + A$. Then

$$D_n(q(\varepsilon)) = \varepsilon^{2M_{\mathrm{ov}}}(C_{\mathrm{ov}} + O(\varepsilon^2)), \tag{9}$$

$$\begin{aligned} C_{\mathrm{ov}} = (-1)^{A(d-1)/2} d^{dA(A+1)/2+A} P_A^{d-1} \operatorname{Disc}(J_{A+1}) \\ \cdot \operatorname{Disc}(U) \operatorname{Res}(F,U)^{2A} g^2. \end{aligned} \tag{10}$$

The coefficient is nonzero, and $\operatorname{ord}_{\Phi_d} D_n = M_{\mathrm{ov}}$. No assertion removes (2) or (8), extends this proof to even d, or gives a positive-characteristic unit conclusion.

4 The exact operator and the all order bound

The classical finite q-binomial identity is

$$\mathcal{P}_n(z;q) = \prod_{j=0}^{n-1}(1+q^j z) = \sum_{k=0}^{n} q^{k(k-1)/2} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k. \tag{11}$$

It is recorded, for example, in Gorodetsky [1, equation (1.5)]. Define diagonal and differential operators on polynomials by

$$T_\omega(z^k) = \omega^{k(k-1)/2} z^k, \qquad \mathcal{L} = z^2 \partial_z^2,$$

and set $R_n(z,h) = T_\omega^{-1} \mathcal{P}_n(z; \omega e^h)$. Comparing coefficients in (11) gives the exact formal identity

$$H_n(z;\omega e^h) = \exp(-h\mathcal{L}/2) R_n(z,h). \tag{12}$$

This takes place in the h-adic module of polynomials in z of degree at most n. The coefficient of a fixed h power uses only finitely many terms of the exponential.

Oddness of d is essential to the next step. The multiplier ratio for z^{k+d} and z^k is

$$\omega^{(k+d)(k+d-1)/2 - k(k-1)/2} = \omega^{dk + d(d-1)/2} = 1.$$

Consequently T_ω and its inverse commute with multiplication by z^d and by every power of F. For even d the same ratio is -1.

Lemma 3. If $R_n(z,h) = \sum_{r \geq 0} h^r R_{n,r}(z)$, then

$$F^{\max(A-r,0)} \mid R_{n,r} \quad \text{for every } r \geq 0, \qquad R_{n,0} = F^A G. \tag{13}$$

Proof. Partition the first Ad factors of $\mathcal{P}_n$ into A consecutive blocks of length d, followed by a residual block of length b. At $h = 0$, each full block equals F. In a term of the rth product derivative, at most r full blocks are differentiated. At least $A - r$ intact blocks remain if $r \leq A$. Apply T_ω^{-1}, which commutes with their product. Division by $r!$ does not affect divisibility. The residual finite q-binomial identity gives $R_{n,0} = F^A G$. This proves (13) at every order. $\square$

At a root α of F, put $u = z - \alpha$ and assign u weight 1 and h weight 2. The rth summand of R_n has weighted order at least

$$2r + \max(A - r, 0). \tag{14}$$

This is at least $A + 1$ for $r \geq 1$ and at least $A + 2$ for $r \geq 2$, including $r > A$.

In these local coordinates,

$$h\mathcal{L} = h\alpha^2 \partial_u^2 + 2h\alpha u \partial_u^2 + h u^2 \partial_u^2. \tag{15}$$

The three terms preserve weight, increase it by one, and increase it by two, respectively. Hence $\exp(-h\mathcal{L}/2)$ preserves every weighted remainder ideal. On a fixed leading weight its action is $\exp(-h\alpha^2 \partial_u^2/2)$. Its weight-preserving part is a finite sum on each weighted homogeneous polynomial, because repeated second derivatives terminate. The full expression is also well defined h-adically by (12). No analytic convergence is used.

5 Proof of the initial forms

5 1 The separated coefficient

Since $F'(\alpha) = -d/\alpha$, the weight-A part of R_n is

$$(-d/\alpha)^A G(\alpha) u^A.$$

All positive-r parts have larger weight by (14). The finite polynomial identity

$$\exp(-s\partial_u^2/2)u^A = \sum_{j=0}^{\lfloor A/2 \rfloor} \frac{A!(-s)^j u^{A-2j}}{2^j j!(A-2j)!} \tag{16}$$

equals $s^{A/2}\,\mathrm{He}_A(u/\sqrt{s})$. Apply (16) with $s = h\alpha^2$, then put $u = \varepsilon x$ and $h = \varepsilon^2/d + O(\varepsilon^4)$. The result is (1). Formula (16) also explains directly why square roots cancel.

5 2 The first product derivative at the shared root

Let $\mathcal{D} = z\partial_z$ and define

$$E_b(z) = \prod_{r=0}^{b-1}(1+\omega^r z),$$

$$S(z) = \sum_{r=0}^{d-1} r\omega^r z \prod_{\substack{0\le s<d \\ s\neq r}} (1+\omega^s z),$$

with S_b defined by the same sum and product over $0 \le r, s < b$. The derivative of full block i, indexed from $i = 0$, is $S + id^2 z^d$. Therefore

$$\begin{aligned} \partial_h \mathcal{P}_n(z;\omega e^h)\big|_{h=0} = F^{A-1}\left(AE_bS + \frac{d^2A(A-1)}{2} z^d E_b\right) \\ + F^A\left(Ad\mathcal{D}E_b + S_b\right). \end{aligned} \tag{17}$$

Apply T_ω^{-1}. The $z^d E_b$ term acquires a factor G, which vanishes at -1. The final F^A term has weight at least $A+2$ after multiplication by h. Thus, with $u = z+1$ and

$$c_b = (T_\omega^{-1}(E_bS))(-1),$$

the only weight-$(A+1)$ contribution from $hR_{n,1}$ is $Ad^{A-1}c_b h u^{A-1}$. This statement also holds for $A = 1$; the term with factor $A(A-1)$ is then zero.

We can determine c_b using $A = 1$. Differentiating (12) gives

$$R_{d+b,1} = \partial_h H_{d+b}(z;\omega e^h)\big|_{h=0} + \frac{\mathcal{L}}{2}(FG).$$

Because $FG = dg\,u^2 + O(u^3)$, the second term equals dg at $z = -1$. The classical Gaussian value is

$$H_{2m}(-1;q) = (q;q^2)_m = \frac{(q;q)_{d+b}}{(q^2;q^2)_m}, \qquad 2m = d+b. \tag{18}$$

Odd-degree values vanish. These identities also follow from the three-term recurrence displayed in Appendix A; see Schultz–Walker [2, printed page 2]. The numerator on the right of (18) contains exactly one vanishing factor at $q = \omega$, namely $1 - q^d$. The remaining factors multiply to $d(\omega;\omega)_b$, since $\prod_{j=1}^{d-1}(1-\omega^j) = d$. Also $\partial_h(1 - e^{dh})|_{h=0} = -d$. Therefore

$$H_{d+b}(-1;\omega e^h) = -d^2Wh + O(h^2), \qquad c_b = dg - d^2W = dg(1-\tau). \tag{19}$$

The weight-$(A+1)$ static term is $d^A g u^{A+1}$, while every $r \geq 2$ term has weight at least $A+2$. The complete initial form of R_n is consequently

$$d^A g\left(u^{A+1} + A(1-\tau)hu^{A-1}\right). \tag{20}$$

At $\alpha = -1$, apply $\exp(-h\partial_u^2/2)$ to (20) and substitute $u = \varepsilon x$, $h = \varepsilon^2/d + O(\varepsilon^4)$. Formula (16) gives (7). Both the factor d in $\tau = dW/g$ and the factor $1/d$ in the leading term of h are necessary.

6 Ordinary discriminants and signs

Under the respective guards, each simple root of the initial polynomial lifts by the formal implicit-function theorem to a root branch in an algebraic closure with coefficients in formal ε series. In the separated case there are A branches at each of the d roots of F, together with b residual branches. In the overlap case there are $A+1$ branches at -1, A at each other root of F, and $b-1$ residual branches. These counts account for all n roots.

6 1 Separated clusters

If $h_1, \ldots, h_A$ are the roots of He_A, the first displacements at α are $\alpha h_j/\sqrt{d}$. The product of squared differences within all clusters is therefore, to leading order,

$$\varepsilon^{dA(A-1)} d^{-dA(A-1)/2} P_A^d. \tag{21}$$

The omitted α powers cancel: $\prod_{F(\alpha)=0}\alpha = -1$ and $A(A-1)$ is even. The cross-cluster product is $\mathrm{Disc}(F)^{A^2}$, where

$$\mathrm{Disc}(F) = (-1)^{(d-1)/2} d^d. \tag{22}$$

Residual pairs and pairs between clusters and residual roots contribute $\mathrm{Disc}(G)\,\mathrm{Res}(F,G)^{2A}$. Multiplying these terms, using $A^2 \equiv A \pmod 2$, proves (3) and (4).

6 2 The shared cluster

The first displacements in (7) are $j_k/\sqrt{d}$, where the j_k are the roots of J_{A+1}. The parity convention with negative argument gives the same set. Put $F_0 = F/(z+1)$. Then

$$F_0(-1) = d, \qquad \mathrm{Disc}(F_0) = (-1)^{(d-1)/2} d^{d-2}.$$

The product of the roots of F_0 is 1. The within-cluster contribution is

$$\varepsilon^{2M_{\mathrm{ov}}} d^{-M_{\mathrm{ov}}} P_A^{d-1}\,\mathrm{Disc}(J_{A+1}). \tag{23}$$

Cross-cluster pairs contribute

$$\operatorname{Disc}(F_0)^{A^2} d^{2A(A+1)} = (-1)^{A(d-1)/2} d^{dA^2+2A}. \tag{24}$$

The residual contribution is $\operatorname{Disc}(U)\operatorname{Res}(F,U)^{2A} g^2$. The extra factor g^2 accounts for the extra root in the shared cluster. Combining (23) and (24) proves (9) and (10).

In both cases, nonvanishing of the displayed coefficient establishes the exact order. If a guard fails, these root-lifting arguments do not certify that order. Although individual root branches need not be even in ε, the discriminant is even because $q(\varepsilon)$ is even. This gives the normalized remainders $O(\varepsilon^2)$ in (3) and (9).

7 Conversion between local parameters

Let (M, C) mean either $(M_{\mathrm{sep}}, C_{\mathrm{sep}})$ or $(M_{\mathrm{ov}}, C_{\mathrm{ov}})$, under its hypotheses, and put $t = q^d - 1$. The formulas above say

$$D_n(q) = t^M (C + O(t)) \quad \text{locally at } q = \omega. \tag{25}$$

Since $t = d\omega^{d-1}(q-\omega) + O((q-\omega)^2)$, the coefficient of $(q-\omega)^M$ is

$$(d\omega^{d-1})^M C. \tag{26}$$

It is generally different from C. Let $Q_d(q) = (q^d - 1)/\Phi_d(q)$. The integral polynomial $D_n(q)$ has the same exact root multiplicity at every primitive dth root by Galois conjugacy. Monic division gives

$$D_n(q) = \Phi_d(q)^M E(q), \qquad E(q) \in \mathbb{Z}[q], \qquad E(\omega) = Q_d(\omega)^M C. \tag{27}$$

These assertions are in characteristic zero. Nonvanishing of C there does not imply that its reduction is a unit in any specified positive characteristic.

8 Classical antecedents and logical dependence

The input (11) is classical. The static identity $H_n(z;\omega) = F(z)^A G(z)$ is the usual root-of-unity factorization; (12)–(13) also derive it directly. The special value (18), the Hermite discriminant, and the root-product discriminant rule are established ingredients. The proof here supplies the block filtration and the first surviving coefficients explicitly.

Schultz–Walker [2, Theorem 1.1, printed page 3] provides a relevant all-order vanishing antecedent. If $\alpha^d = -1$ has order $2c$, then $c \mid d$ and d/c is odd. In their theorem take $k = d/c$, their auxiliary exponent and congruence-order parameter equal to zero, derivative index r, and weight polynomial $P(s) = s(s-1)\cdots(s-\ell+1)$. Up to the nonzero factor $\alpha^{-\ell}$, their sum is $\partial_q^r \partial_z^\ell H_n(\alpha; q)$. Their bound becomes $n \geq (\ell + 2r + 1)d$. Hence

$$\partial_q^r \partial_z^\ell H_n(\alpha;\omega) = 0 \quad \text{whenever } 2r + \ell < A. \tag{28}$$

Their primitive-choice passage is on printed page 13. The stated implication covers roots of smaller order as well as primitive $2d$th roots. The cited theorem gives vanishing; it does not specify the surviving coefficient.

The logical distinctions matter. For $b = 0$ and $A \geq 2$, adding $\lambda(q^d - 1)F^{A-2}$ to a monic polynomial family preserves its static specialization and all below-weight-A vanishings, but changes its weight-A initial form. At the shared root, adding $\lambda(q^d - 1)F^{A-1}$ similarly changes weight $A + 1$ while preserving the lower vanishings. These are logical counterexamples to determining a coefficient from vanishing data alone. The perturbed families are not asserted to be Rogers–Szegő polynomials.

Conversely, a full local initial-form identity implies its lower-weight mixed-derivative vanishings. The coordinate change $q - \omega = \omega h + O(h^2)$ is invertible and filtered; substitution $u = \varepsilon x$, $h = \varepsilon^2/d + O(\varepsilon^4)$ separates the coefficients of each weight by powers of x. This local converse does not recover the full Schultz–Walker theorem. Likewise, the guarded initial forms imply the discriminant coefficients, but a single discriminant coefficient does not determine all initial polynomials or imply the operator identity.

The accepted implication is therefore

$$\begin{gathered}\text{finite product identity and proved block filtration} \\ + \text{ shared root special value when needed} \\ \Longrightarrow \text{ local initial forms} \\ \Longrightarrow \text{ guarded discriminant formulas.}\end{gathered}$$

This derivation settles the previously unworked all-A implication route. It does not settle when the resulting formulas were first published. The finite controls in Appendix A support the audit and cannot establish the all-A claim by themselves.

9 Remaining comparisons and extensions

Historical comparison still requires the exact six-page body of Zhedanov's 1997 paper [4], the printed page 360 image of Lubinsky–Saff [3], and substantive bodies along the relevant Lubinsky–Saff citation chain. Missing or inaccessible text supplies no evidence that a theorem is absent. The comparison of the complex-parameter disk and inverse-pair applications is a separate open priority question.

The local theorem itself retains several boundaries. Multiple common residual roots, a repeated root at -1, or a repeated root of J_{A+1} would require a different initial-form analysis. The proof does not remove the composite residual squarefreeness or resultant guards. Even d changes the periodicity used in (13). Uniform estimates when degree grows, and arithmetic conclusions requiring positive-characteristic units, need additional arguments. None of these extensions is claimed here. Appendix C supplies conservative bounds for each specified finite instance; it does not supply efficient or uniform growing-degree estimates, a numerical implementation certificate, or any change in historical-priority status.

For $A = 1$, the separated discriminant is already the static product-discriminant consequence; the simple-overlap calculation reduces to (18) and a Taylor expansion. For arbitrary A, the accepted result is a fully written all-order coefficient derivation under the guards. Any assessment of its publication contribution should separate correctness, explicit derivability, historical priority, and the value of particular applications.

References

[1] Ofir Gorodetsky. *q-congruences, with applications to supercongruences and the cyclic sieving phenomenon.* arXiv:1805.01254v2, equation (1.5), preprint page 2. The citation records the classical finite q-binomial identity and does not assign its origin to this paper. Primary preprint

[2] Andrew Schultz and Robert Walker. *A generalization of the Gaussian formula and a q-analog of Fleck's congruence.* arXiv:1202.0199v1, printed pages 2–3 and 13. Published in *Journal of Number*

Theory 133(11) (2013), 3717–3738, DOI 10.1016/j.jnt.2013.05.003. Page locators here refer to the preprint. Primary preprint

[3] D. S. Lubinsky and E. B. Saff. *Convergence of Padé approximants of partial theta functions and the Rogers–Szegő polynomials* (1987). Historical comparison remains incomplete at the source scope stated in Section 9. Publisher record

[4] A. Zhedanov. *Gauss sums and orthogonal polynomials* (1997). The exact body remains an outstanding comparison source. Publisher record

Appendix A Finite controls and reproducibility

The independent review used a freshly written three-term recurrence implementation:

$$H_0 = 1, \quad H_1 = 1 + z, \quad H_n = (1 + z)H_{n-1} + (q^{n-1} - 1)zH_{n-2}.$$

It computed exact rational truncated ε series in $\mathbb{Q}[w]/(w^d - 1)$, then reduced modulo $\Phi_d(w)$. It did not generate the tested jets using the operator proof or the author's q-Pascal program. The author's code was read as inert text during that review, not imported or executed.

All 16 saved moving-root controls passed. In the table, s specifies the tested root $\alpha = -\omega^s$. "Other root" means evaluation of formula (1); it does not claim that every residual guard holds globally for that parameter triple.

A	d	b	s	Local formula	Initial weight
1	3	1	0	Shared root	2
2	3	2	2	Other root	2
4	3	1	0	Shared root	5
6	3	1	0	Shared root	7
10	3	1	0	Shared root	11
12	3	0	2	Other root	12
5	5	1	0	Shared root	6
4	5	3	0	Shared root	5
3	5	4	1	Other root	3
3	7	5	0	Shared root	4
4	7	2	3	Other root	4
3	9	7	0	Shared root	4
4	9	2	3	Other root	4
2	15	3	5	Other root	2
1	15	13	0	Shared root	2
7	1	0	0	Classical endpoint	7

Five direct ordinary-discriminant computations also passed the exact cyclotomic valuation and both coefficient conversions:

(A, d, b)	Degree n	Exact Φ_d order
$(1, 3, 2)$	5	0
$(1, 5, 1)$	6	1
$(2, 3, 2)$	8	3
$(2, 3, 1)$	7	5
$(1, 5, 3)$	8	1

Six mutation controls detected the intended errors: omitting the heat term; replacing τ by 1; omitting d from τ; assuming the same periodicity for even d; omitting the $q-\omega$ scaling; and dropping the separated resultant guard. Constant-polynomial discriminants were explicitly set to 1.

This appendix reports the sealed independent-control run. The full exact outputs, executable source, and run log are retained in the independent review archive identified below. They test bounded examples; the proof for every A is Lemma 3 and Sections 5–7.

Appendix B Source identities and review boundary

The original proof archive and independent review archive retain their existing identities. This working note is a separate document, not a replacement or a retrospective edit of either archive. The prior independent acceptance concerns the exact proof bytes below; editorial consistency of this note must be distinguished from a new independent mathematical audit.

All order implication archive

F020_All_Order_Implication_Provisional.zip

Library identity: libfile_f90f326d1b0c8191a838d4cb2e7fab76, version 0.

SHA256: `79273ecedccd68bf0d174c843eb5cecc9d32cbb596b115519d26da11352b9ffd`

Accepted proof member ALL_ORDER_IMPLICATION.md

SHA256: `8ca570c7e7c41c50827b1dea3feb931cbcfe0fdab3c3a2016c6e7d0989126abc`

Independent review archive

F020_All_Order_Independent_Review.zip

Library identity: libfile_770d307623188191a6408f4915be0687, version 0.

SHA256: `45671b26ad19d741ca9a8b08da11db23d946a60f6758f18b98544b2a1a8f860e`

Independent review member INDEPENDENT_REVIEW.md

SHA256: `aa6fab3492f265fca7a667a213f57f1453dcfb280d94280ce74b330b353bffd3`

Independent control output INDEPENDENT_CONTROLS.json

SHA256: `778d7585963a6bef193280895da256cc6cac6ee6c6b5ab95e8a2f8ac0216f0ab`

Independent control source independent_controls.py

SHA256: `2ddd2cbef7c652152aae23f7877494bec320ff6c4e8ec4e9c56cd2847cca14a0`

The independent verdict accepts the characteristic-zero all-A implication in the separated and one-simple-overlap scope, with no correction required to its displayed formulas. It checked selected primary PDF text for [1] and [2]. Its attempted primary page-image retrieval failed; it does not claim a rendered-page inspection or an exhaustive whole-paper negative result.

For the earlier separated audit, the recovered file has the following SHA256.

`f093175f18d1941a99a605a5d816b1db31e5b2c4161ee7df4bf29291ff8eff3c`

The catalogue lists a distinct export SHA256.

`4b86b3078bbdf644a3317e67449a974c356e8b2dd8a9f66a393df71341d3e8d9`

It is not silently identified with those bytes. The original separated and overlap reports, and their independent acceptance records, remain unchanged.

Appendix C Finite parameter analytic certificates

This appendix turns the accepted initial forms into explicit analytic error and separation bounds for fixed finite parameters. Its source certificate received an independent mathematical acceptance on 11 October 2026. The argument is a conservative application of Cauchy's estimate and Rouché's theorem. It makes no claim of optimal constants, efficient growing-degree behavior, implementation stability, or historical novelty. Appendix D binds this integration to the accepted source bytes.

C 1 Analytic branch and normalized remainder

Fix a complex embedding of the data in Section 2, with odd $d \geq 3$, $A \geq 1$, and $0 \leq b < d$. Retain either the separated guard (2) or all the simple-overlap hypotheses of Section 3.2, including odd b and (8). On $|\varepsilon| < 1$, use

$$q(\varepsilon) = \omega \exp\left(\frac{\operatorname{Log}(1+\varepsilon^2)}{d}\right).$$

The principal logarithm is analytic here because $\operatorname{Re}(1+\varepsilon^2) > 0$. Its Taylor series agrees with the formal branch in Section 2. Every root α of F has $|\alpha| = 1$. Write its initial form as

$$H_n(\alpha + \varepsilon x; q(\varepsilon)) = \varepsilon^{m_\alpha}(P_\alpha(x) + O(\varepsilon)). \tag{C1}$$

At an ordinary center, $m_\alpha = A$ and P_α is the coefficient in (1); at the exceptional center -1, $m_{-1} = A+1$ and P_{-1} is the coefficient in (7). If c_α denotes the leading coefficient, then

$$|c_\alpha| = d^A|G(\alpha)| \quad \text{at an ordinary center,} \qquad |c_{-1}| = d^A|g| \quad \text{at the exceptional center.}$$

All these P_α have simple roots under the retained guards.

For $Q \geq 1$ and $Z \geq 0$, define the positive majorant

$$\mathcal{B}_n(Q, Z) = \sum_{k=0}^{n} \binom{n}{k} Q^{k(n-k)} Z^k \leq Q^{\lfloor n^2/4 \rfloor}(1+Z)^n, \qquad Q_s = (1+s^2)^{1/d}. \tag{C2}$$

For $X \geq 0$, $0 < r < s < 1$, $0 < |\varepsilon| \leq r$, and $|x| \leq X$, the explicit normalized remainder is

$$|\varepsilon^{-m_\alpha} H_n(\alpha + \varepsilon x; q(\varepsilon)) - P_\alpha(x)| \leq E_\alpha(r; s, X) := \frac{\mathcal{B}_n(Q_s, 1+sX)}{s^{m_\alpha}} \frac{r}{s-r}. \tag{C3}$$

Indeed, the Gaussian polynomial has nonnegative integer coefficients, coefficient sum $\binom{n}{k}$, and degree $k(n-k)$, by its polynomial Pascal recurrence. On $|\varepsilon| = s$, its contribution is bounded by (C2). Cauchy's estimate bounds the ℓth coefficient of the unnormalized function by $\mathcal{B}_n(Q_s, 1+sX)s^{-\ell}$. Equation (C1) cancels the coefficients below m_α; summing the tail proves (C3). In particular the normalization factor s^{-m_α} is essential.

C 2 Certified moving root tubes

Write $P_\alpha(x) = c_\alpha \prod_{j=1}^{m_\alpha}(x - \xi_{\alpha,j})$. For $m_\alpha > 1$, let δ_α and D_α be the minimum and maximum pairwise direction distances, and choose $0 < \eta_\alpha < \delta_\alpha/2$. For $m_\alpha = 1$, choose any $\eta_\alpha > 0$. Set

$$X_\alpha = \max_j |\xi_{\alpha,j}| + \eta_\alpha, \qquad \mu_\alpha = |c_\alpha|\eta_\alpha \min_j \prod_{k \neq j}(|\xi_{\alpha,j} - \xi_{\alpha,k}| - \eta_\alpha). \tag{C4}$$

The empty product for a singleton is one. If

$$E_\alpha(r; s, X_\alpha) < \mu_\alpha, \quad \text{equivalently} \quad r < \frac{\mu_\alpha s^{m_\alpha+1}}{\mathcal{B}_n(Q_s, 1+sX_\alpha) + \mu_\alpha s^{m_\alpha}}, \tag{C5}$$

then for every $0 < |\varepsilon| \le r$, each disk

$$|z - (\alpha + \varepsilon\xi_{\alpha,j})| < |\varepsilon|\eta_\alpha \tag{C6}$$

contains exactly one zero, counted with multiplicity, of H_n. It is therefore a simple zero. For two roots in this cluster,

$$|\varepsilon|(\delta_\alpha - 2\eta_\alpha) < |z_i - z_j| < |\varepsilon|(D_\alpha + 2\eta_\alpha). \tag{C7}$$

To prove this, the product in (C4) bounds $|P_\alpha|$ from below on each circle $|x - \xi_{\alpha,j}| = \eta_\alpha$. Rouché's theorem and (C3) preserve the single-zero count. Rescaling by the nonzero complex number ε proves (C6), and the triangle inequality proves (C7). The bound in (C5) is positive and less than s, so every guarded finite instance admits a nonzero local radius.

An explicit quadratic root error is also available. Fix an admissible η_0, put $X_0 = \max_j |\xi_{\alpha,j}| + \eta_0$, and set

$$L_\alpha = |c_\alpha| \min_j \prod_{k\neq j} (|\xi_{\alpha,j} - \xi_{\alpha,k}| - \eta_0).$$

If $2E_\alpha(r; s, X_0)/L_\alpha \le \eta_0$, apply (C5) with $\eta_\alpha = 2E_\alpha(r; s, X_0)/L_\alpha$: its boundary lower bound is at least twice the error. Thus

$$|z_j - (\alpha + \varepsilon\xi_{\alpha,j})| < \frac{2|\varepsilon|E_\alpha(r; s, X_0)}{L_\alpha}. \tag{C8}$$

For $r = |\varepsilon|$ and fixed s, X_0, L_α, this is an explicit $O(|\varepsilon|^2)$ remainder.

C 3 Residual roots and the exact Q08 interface

Let a run over the distinct roots of $f_0 = F^A G$, with multiplicity m_a. A residual center β is a simple root of G in the separated case, or of U in the overlap case. Choose $0 < \rho_\beta < \min_{a\neq\beta} |\beta - a|$ and define

$$\nu_\beta = \rho_\beta \prod_{a\neq\beta} (|\beta - a| - \rho_\beta)^{m_a}.$$

The static polynomial is monic, so $|f_0| \ge \nu_\beta$ on the circle about β. At fixed z, the moving polynomial is even in ε. The even Cauchy tail therefore gives

$$|H_n(z; q(\varepsilon)) - f_0(z)| \le \mathcal{B}_n(Q_s, |\beta| + \rho_\beta) \frac{r^2}{s^2 - r^2} < \nu_\beta. \tag{C9}$$

Whenever the final strict inequality holds, Rouché certifies one simple root in $|z - \beta| < \rho_\beta$. Its analytic branch is even and equals $\beta + O(\varepsilon^2)$, since $f_0'(\beta) \neq 0$.

Use one common r, s in all the local tests. Give each moving cluster the static-center radius $R_a = rX_a$, and each residual singleton the radius $R_\beta = \rho_\beta$. Require also

$$\Delta = \min_{a\neq c}(|a - c| - R_a - R_c) > 0, \qquad R = \max_a(|a| + R_a). \tag{C10}$$

The certified roots then exhaust the degree: their counts are $dA + b$ in the separated case and $(d - 1)A + (A + 1) + (b - 1) = n$ in the overlap case. Distinct-cluster distances are at least Δ, and all nodes have modulus at most R. For moving clusters with $m_a > 1$, put $\gamma_a = \delta_a - 2\eta_a > 0$.

For these n nodes define the square moment Vandermonde $V_{k+1,i} = z_i^k$, $0 \le k < n$, $1 \le i \le n$. The already accepted Q08 finite-separation envelope gives

$$\|V^{-1}\|_2 \le \sqrt{n}(1 + R)^{n-1} \max_a \left\{ (|\varepsilon|\gamma_a)^{-(m_a-1)} \Delta^{-(n-m_a)} \right\}, \qquad \|V\|_2 \le \sqrt{n \sum_{k=0}^{n-1} R^{2k}}. \tag{C11}$$

Here $\|\cdot\|_2$ is the Euclidean operator norm. For every singleton the first factor in braces is defined to be one; no γ_a is needed. The bound follows row by row from Lagrange interpolation: numerator coefficient ℓ^1 norm is at most $(1+R)^{n-1}$, and its denominator has m_a-1 within-cluster factors and $n-m_a$ cross-cluster factors. The inverse Frobenius norm supplies $\sqrt{n}$. The second inequality is the Frobenius bound for V.

Q08 separately already proves the fixed-degree singular-value exponents $\bigsqcup_a\{0,\dots,m_a-1\}$, sorted in increasing order for singular values in decreasing order. All comparison constants may depend on the degree and geometry. Equations (C3)–(C10) supply verified inputs to (C11); they are not a new singular-value-profile theorem. Root preparation and rigorous enclosures still have a cost. Resolvent legality, including $1-\lambda\zeta_N^k\neq 0$ for $\lambda=-z$, $\zeta_N=e^{2\pi i/N}$, positive integer N, and $0\le k<N$, is a separate guard. A varying q here is not a continuous family of primitive roots of one fixed order.

C 4 A parameter only separated subfamily

For $b=0$ and $A\ge 2$, the Hermite directions obey

$$\sqrt{\frac{A-1}{d}}\le\max_j|\xi_{\alpha,j}|\le 2\sqrt{\frac{A-1}{d}},\qquad \frac{\pi}{\sqrt{d(A+1/2)}}\le\delta_\alpha\le\frac{4}{\sqrt{d(A-1)}}. \tag{C12}$$

For completeness, the Hermite recurrence realizes its zeros as the simple real eigenvalues of the symmetric tridiagonal matrix with off-diagonals $\sqrt{1},\dots,\sqrt{A-1}$. Its trace square is $A(A-1)$, while its row-sum norm is at most $2\sqrt{A-1}$. The smallest gap is at most the total span divided by $A-1$. For the lower gap, $u=e^{-y^2/4}\,\mathrm{He}_A(y)$ satisfies $u''+(A+1/2-y^2/4)u=0$. On any interval between consecutive zeros, integration by parts and the Dirichlet Poincaré inequality give an interval length at least $\pi/\sqrt{A+1/2}$. Scaling by $1/\sqrt{d}$ proves (C12). The recurrence and differential equation are classical; see NIST DLMF Tables 18.9.1 and 18.8.1.

Put

$$\delta_0=\frac{\pi}{\sqrt{d(A+1/2)}},\quad \eta=\delta_0/4,\quad X=2\sqrt{(A-1)/d}+\eta,\quad \mu=d^A\eta(\delta_0-\eta)^{A-1}.$$

For fixed $0<s<1$, let $B=\mathcal{B}_{Ad}(Q_s,1+sX)$. All Ad roots lie in the disjoint tubes if

$$0<|\varepsilon|<\min\left\{\frac{\mu s^{A+1}}{B+\mu s^A},\frac{\sin(\pi/d)}{X}\right\}. \tag{C13}$$

The within-cluster lower bound is $|\varepsilon|\delta_0/2$, and the cross-cluster lower bound is $2\sin(\pi/d)-2|\varepsilon|X>0$. This is (C5) with $|c_\alpha|=d^A$ and the exact static-center gap $2\sin(\pi/d)$. It provides a nonempty explicit domain for each finite pair, not an efficient degree-uniform regime. The geometric condition suggests small $|\varepsilon|\sqrt{Ad}$ as parameters vary; it is only a sufficient disk condition, not a necessary collision threshold. The Cauchy condition may be much more restrictive.

C 5 Exact finite examples

Both examples use $s=1/2$, $r=1/10000$, $\eta=1/8$, and the convenient disk bound $X=1$, which is at least each required X_α. All inequalities below hold for complex $0<|\varepsilon|\le r$.

Separated example. For $(A,d,b)=(2,3,0)$, each cluster has $|c|=9$, directions $\pm\alpha/\sqrt{3}$, direction gap >1, and maximum direction modulus $<3/5$. Thus $\mu\ge 63/64$, while (C2)–(C3) give

$$B\le(5/4)^3(5/2)^6=\frac{1953125}{4096},\qquad E\le\frac{1953125}{5118976}<\frac{63}{64}. \tag{C14}$$

The three static centers have gap $\sqrt{3}>1$; their radius-r envelopes are disjoint. All six roots are certified, and within-cluster distances exceed $3|\varepsilon|/4$.

Simple-overlap example. For $(A, d, b) = (1, 3, 1)$, $U = g = 1$ and $\tau = 3/(1-\omega^2) = 1-\omega$, so $|\tau| = \sqrt{3}$. At -1, $P(x) = 3x^2 - \tau$ has direction modulus $3^{-1/4} < 7/8$, gap > 1, and $\mu \geq 21/64$. The other two clusters are linear, with $|c| = 3\sqrt{3}$ and $\mu > 3/8$. A common deliberately coarse majorant is $B \leq (5/4)^4(5/2)^4 = 390625/4096$. Consequently

$$E_{-1} \leq \frac{390625}{5118976} < \frac{21}{64}, \qquad E_{\text{ordinary}} \leq \frac{390625}{10237952} < \frac{3}{8}. \tag{C15}$$

All four roots are certified in disjoint static-center envelopes.

The independent finite-parameter review reconstructs these local jets exactly in $\mathbb{Q}[w]/(w^2+w+1)$, using weighted-subset Gaussian coefficients. Its 91 Gaussian-coefficient controls through $n = 12$, tail algebra, Hermite identities through $A = 24$, Lagrange-envelope control with heterogeneous clusters and a singleton, and discriminant threshold controls all pass. They support the proof, rather than prove the general statement by extrapolation. At $r = 1/1000$ in the separated example, the correct sufficient test is inconclusive, while omitting s^{-m_α} makes an incorrect test pass. This detects a normalization error; it is not evidence of an actual root collision.

C 6 Discriminant truncation error

Set $t = \varepsilon^2$ and $q(t) = \omega \exp(\mathrm{Log}(1+t)/d)$ for $|t| < 1$. Use $(M, C) = (M_{\text{sep}}, C_{\text{sep}})$ or $(M_{\text{ov}}, C_{\text{ov}})$ from Section 3, so $D_n(q(t)) = t^M(C + O(t))$ and $C \neq 0$. For $0 < v < 1$, put

$$S = \mathcal{B}_n((1+v)^{1/d}, 1), \qquad T = n^n S^{2n-1}.$$

On $|t| = v$, the coefficient ℓ^1 norms of H_n and its ordinary z-derivative are at most S and nS. The Sylvester matrix has $n-1$ shifted polynomial rows and n shifted derivative rows. Hadamard's inequality bounds its determinant in modulus by T, and hence bounds the monic discriminant by T. The normalized Cauchy tail yields, for $0 < |t| < v$,

$$|t^{-M} D_n(q(t)) - C| \leq T v^{-M} \frac{|t|}{v - |t|}. \tag{C16}$$

For any specified relative tolerance $\theta > 0$, the strict condition

$$|t| < \frac{\theta |C| v^{M+1}}{T + \theta |C| v^M} \tag{C17}$$

makes this error less than $\theta|C|$. This includes $M = 0$. It certifies truncation error only. It does not certify accurate floating-point evaluation of a large discriminant or infer the individual singular spectrum from a discriminant alone.

Appendix D Revision and source provenance

This is a separate Version 2 working note. The original PDF and LaTeX, the accepted all-order proof, and both finite-parameter source archives are preserved unchanged in the companion sealed package. Sections 2–8 and Appendices A–B are unchanged. The only earlier-body edits are the version label, the introductory description of this addition, and the finite-parameter clarification in Section 9. Paragraph spacing is adjusted for clean pagination. The package manifest binds all retained sources, this revised PDF and LaTeX, and verification files.

The independent source verdict is *ACCEPT* for the finite-parameter analytic remainder, root tubes, complete separation, and discriminant truncation bounds. It requires no mathematical correction. This acceptance concerns its exact frozen source bytes. The present integration is an editorial revision; independent document-fidelity review of it is a separate step. No historical-priority, implementation, or research-ranking upgrade follows from this addition.

Original working note. PDF Library identity libfile_72ddfda84adc8191b5ed994749139490; LaTeX identity libfile_46168cfa2b3481918766ba14fbc98d66, both version 0. Their SHA256 values, respectively, are

c65f50a5dc3acc0580f35e17ead1adbd9b5dc6a7d7f42d4970e8e1c3809e9372

23462defbe98daf588869a27e46f4ad94f67fd02a09e0196d31bf231aa15c1f0

Accepted finite-parameter source. Archive Library identity libfile_730b9b6c2d708191bc51cf5419553afe, version 0. Archive SHA256:

2f2180470328e69db523339981b83918698b62f2589ad750010806b4fc8923ef

The accepted LaTeX member SHA256 is

3b472a7adb4a157c52c6ad4f8da5e1d94717e360bce8326e8c4d8732cfbcc100

Independent finite-parameter review. Archive Library identity libfile_30c794c5011c819196acc7d3104d8aad, version 0. Archive SHA256:

b5a00e74f00f1f9d2b5ebfc1afb9479bb950efca792504d63a1c79ce6ffc3e72

The accepted review member SHA256 is

a412af68fbeffe1d6f332280a5b59e3ac74d91481c492232981319c51e0b6033

Q08 interface dependencies. The source *Structured fusion second pass*, Section S3, supplies the fixed-degree singular profile and finite inverse envelope. Its SHA256 is

02ec40df49f0585c57f98d3af7cbb9867216773f98a2998db9d51237fd69b787

The separate resolvent and root-path warning is in *Fusion proofs*, Theorem T3, SHA256

4afd27bacea1bd896d8b44e7e4c1eb51c1b24b3db997ceca9e554e69f81202bb

The accepted all-order proof and review are pinned in Appendix B. Unverified approximate root lists are not certificates: numerical use of Appendix C requires exact algebraic data or rigorous outward-rounded enclosures. The historical-source limitations in Section 9 remain in force.